Lab-Manual - Programme: B.Tech. (CSE)

Semester: VI

Course Title: Cloud Computing Lab

Course Code: SOE-B-CSE-23-606

Credits: 1 | Contact Hours: 2 Hrs / Week | Max Marks: 50

Course Description

This laboratory course provides hands-on experience with the core concepts and practical applications of cloud computing. Students will explore leading cloud platforms such as Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP) to understand cloud infrastructure, deployment models, and service models. The course emphasizes practical exposure to virtualization, cloud storage, server less computing, cloud databases, identity management, DevOps tools, and deployment of scalable web applications.

Course Outcomes (COs)

At the end of the course, students will be able to:

CO No.	Course Outcome
CO1	Provision virtual machines, implement storage solutions, configure auto-scaling and load balancing, and deploy applications using platform services
CO2	Apply modern DevOps tools and practices including containerization
CO3	Develop, deploy, and manage applications in a cloud environment

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List of Experiments

Experiment 1: Introduction to Cloud Platforms

Aim: To understand cloud platforms and set up cloud accounts.

Tools/Platforms: AWS Free Tier / Azure Student Account / GCP Free Tier

Procedure:

2. Log in to the cloud management console.
3. Explore services such as compute, storage, and networking.

Outcome: Students become familiar with cloud dashboards and services.

Experiment 2: Launching and Configuring Virtual Machines

Aim: To create and configure a virtual machine in the cloud.

Tools/Platforms: AWS EC2 / Azure Virtual Machines

Procedure:

1. Launch an EC2 instance or Azure VM.
2. Configure security groups and key pairs.
3. Connect to the VM using SSH.
4. Install required software packages.

Outcome: Ability to deploy and manage cloud virtual machines.

Experiment 3: Working with Object Storage

Aim: To understand cloud-based object storage services.

Tools/Platforms: AWS S3 / Azure Blob Storage

Procedure:

1. Create a storage bucket/container.
2. Upload files to cloud storage.
3. Retrieve and delete stored objects.

Outcome: Knowledge of cloud storage operations.

Experiment 4: Creating and Deploying a Website

Aim: To host static and dynamic websites on cloud platforms.

Tools/Platforms: AWS S3 / Azure Blob Storage, AWS Elastic Beanstalk / Azure App Service

Procedure:

2. Deploy a dynamic web application using Elastic Beanstalk or App Service.

Outcome: Ability to deploy web applications on the cloud.

Experiment 5: Load Balancing and Auto Scaling

Aim: To implement load balancing and auto-scaling mechanisms.

Tools/Platforms: AWS Elastic Load Balancer, Auto Scaling Groups

Procedure:

1. Configure a load balancer.
2. Create auto-scaling rules based on CPU utilization.

Outcome: Understanding scalable and fault-tolerant cloud applications.

Experiment 6: Cloud Database Services

Aim: To create and use cloud database services.

Tools/Platforms: Amazon RDS / Azure SQL Database

Procedure:

1. Create a cloud database instance.
2. Connect the database with an application.

Outcome: Ability to manage cloud-hosted databases.

Experiment 7: Server less Computing

Aim: To develop and deploy server less applications.

Tools/Platforms: AWS Lambda, API Gateway / S3

Procedure:

1. Create a Lambda function.
2. Trigger the function using API Gateway or S3 events.

Outcome: Understanding server less computing concepts.

Experiment 8: Identity and Access Management (IAM)

Aim: To manage users and permissions in the cloud.

Tools/Platforms: AWS IAM / Azure Active Directory

Procedure:

1. Create IAM users and roles.
2. Assign permissions and policies.
3. Test access control.

Outcome: Knowledge of cloud security and access management.

Experiment 9: Virtualization with Virtual Box / Docker

Aim: To understand virtualization and containerization.

Tools/Platforms: Virtual Box / Docker

Procedure:

1. Install Virtual Box or Docker.
2. Create and run virtual machines or containers.

Outcome: Practical exposure to virtualization and containerization.

Experiment 10: CI/CD in the Cloud (Optional)

Aim: To automate application deployment using CI/CD tools.

Tools/Platforms: AWS Code Pipeline / GitHub Actions

Procedure:

1. Set up a basic CI/CD pipeline.
2. Automate deployment of a simple application.

Outcome: Understanding DevOps practices in cloud environments.

CO–Experiment Mapping

Experiment	CO1	CO2	CO3
1	✓	–	–
2	✓	–	✓
3	✓	–	✓
4	✓	–	✓
5	✓	–	✓
6	✓	–	✓
7	✓	–	✓
8	–	–	✓
9	–	✓	✓
10	✓	✓	✓

Assessment Scheme

- Continuous Lab Evaluation
- Lab Record and Viva Voce

Text Books

1. Anthony T. Velte, Toby J. Velte, Robert Elsenpeter, *Cloud Computing: A Practical Approach*, Tata McGraw-Hill.
2. John W. Rittinghouse, James F. Ransome, *Cloud Computing: Implementation, Management and Security*, CRC Press.

Reference Books

1. Rajkumar Buyya et al., *Cloud Computing: Principles and Paradigms*, Wiley.
2. Ronald L. Krutz, Russell Dean Vines, *Cloud Security*, Wiley.

Tools & Platforms

- AWS Educate / Free Tier
- Microsoft Azure for Students
- Google Cloud Platform (GCP) Free Tier
- Docker, VirtualBox